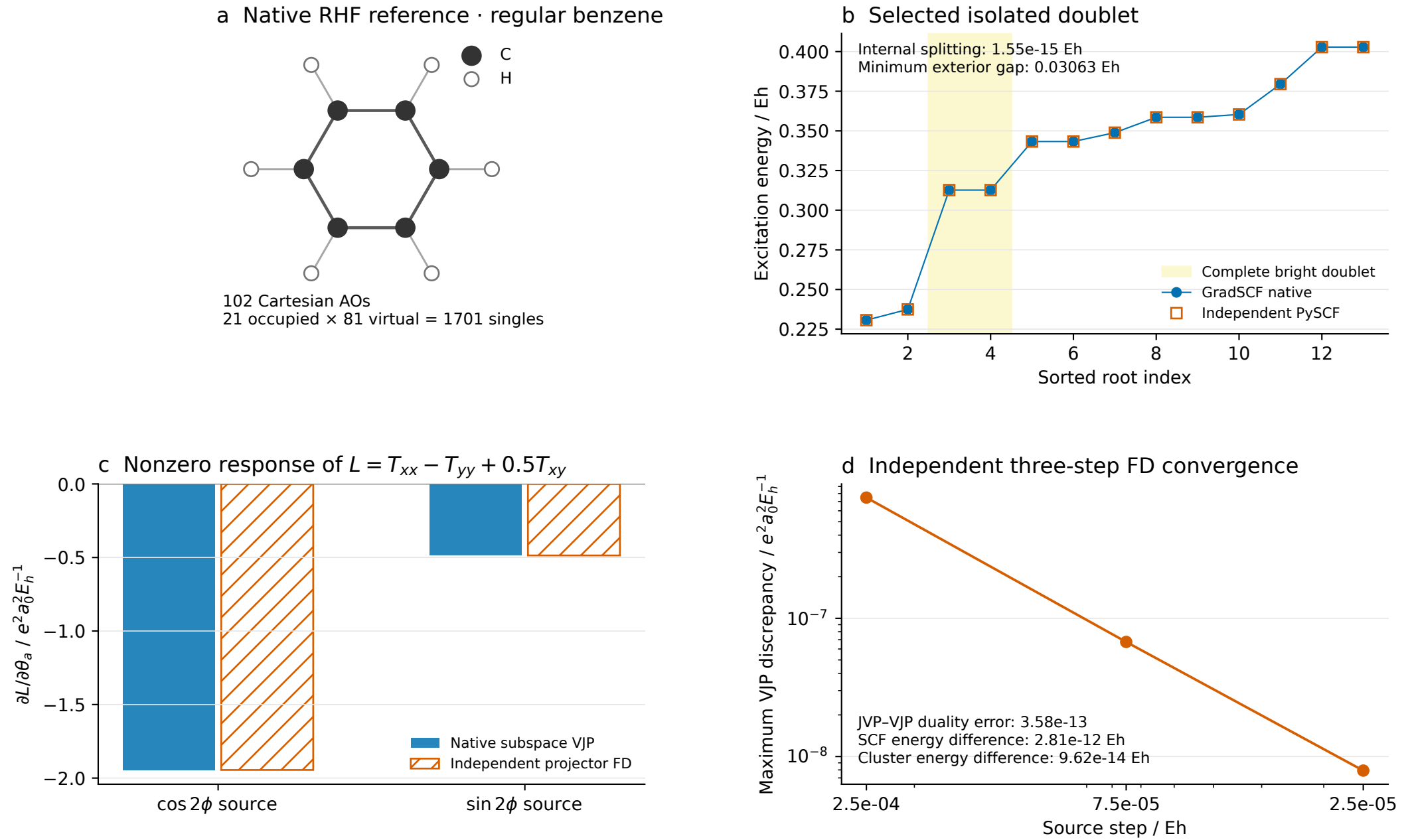
Benzene RHF/CIS · complete degenerate-subspace VJP · Cartesian 6-31G*
GradSCF native SCF/integrals and shared implicit response; independent PySCF SCF/projector differences



$T = D^T P_{\text{doublet}} D$; both members are retained through the symmetry-breaking source.
$V_a = (SW_a + W_aS)/2$ is an inversion-even atomic AO source. Orbitals/integrals are fixed during the response.
This validates the fixed-reference projected CIS model. Relaxed HF/geometry response and adjoint residual telemetry are not included.